

Steps for Running BlueOS on Raspberry Pi

1. Download the BlueOS Image and Flash the MicroSD Card

Start by obtaining the correct operating system image for your Raspberry Pi hardware. The ARMv7 architecture image covers models up to the Raspberry Pi 4, while the ARMv8/ARM64 image is utilized for the Raspberry Pi 5.

- **Prep the SD Card:** A fresh MicroSD card with at least 32GB of capacity is highly recommended. While smaller cards can boot the OS, the extra storage is essential for onboard data recording and buffering high-bandwidth video streams.
- **Flashing:** Open BalenaEtcher, select the downloaded OS image file, choose your target MicroSD card, and flash it.

Relevant Resources:

- Download the latest OS image from the [BlueOS GitHub Releases page](#).
- Download the flashing software from the [BalenaEtcher website](#).

2. Boot the OS and Establish a Direct Connection

Insert the flashed SD card into your Raspberry Pi and apply power. The initial boot usually takes about two minutes because the operating system needs to expand its filesystem to utilize the full capacity of your SD card.

- **Wired Connection:** Connecting the Pi directly to your computer's Ethernet port provides the most stable connection for initial configuration.
- **Wireless Hotspot Fallback:** If BlueOS does not detect a familiar Wi-Fi network within 5 minutes of booting, it will automatically start broadcasting its own wireless hotspot. If you see the "BlueOS" network pop up in your Wi-Fi settings, the correct password to connect is **blueosap**.

3. Access the Web Server Interface

BlueOS operates entirely as a headless system. You will not use an HDMI monitor; instead, you manage the system through a web dashboard.

- **Navigating to the Dashboard:** Open your web browser and type in `http://blueos.local` or `http://blueos-avahi.local`. If you are wired in via Ethernet and the `.local` address isn't resolving, you can access the interface directly using the default IP address: `192.168.2.2`.
- **Internet Access:** Once you are in the interface, click the Wi-Fi indicator in the header to connect the Pi to your local internet network. This allows the system to pull down the latest software updates and extension registries.

Relevant Resource:

- Read through the network troubleshooting steps in the [BlueOS Official Documentation](#).

4. Explore Features, Telemetry, and Extensions

The dashboard acts as the central command hub for hardware interfacing and system logic.

- **Video Streams:** Plug your USB camera into the Pi. BlueOS automatically detects H.264-capable cameras and begins streaming the feed. You can configure multiple camera inputs and adjust stream settings in the Video Streams menu, which is the foundational step before feeding that visual data into photogrammetry software.
- **Control Systems and Telemetry:** Navigate to the Autopilot Parameters and MAVLink Endpoints pages. This section allows you to interface with connected flight controllers, route telemetry data, and fine-tune variables like your PID control loops.
- **Extensions:** BlueOS supports a robust ecosystem of Docker-based extensions. You can browse the BlueOS Bazaar within the interface to install community-built tools, or write your own custom extensions to handle specialized photogrammetry processing directly onboard.

Relevant Resource:

- Learn how to write and deploy your own software tools in the [BlueOS Extensions Documentation](#).